

$^9\text{Be}(^{37}\text{Ca}, ^{34}\text{Ar}\gamma)$ 2007DoZV

$^{37}\text{Ca} \rightarrow 2\text{p}1\text{n} + ^{34}\text{Ar}$ on ^9Be target.

2007DoZV: a 420-MeV/nucleon ^{40}Ca primary beam was provided from the heavy ion synchrotron SIS at GSI and impinged on a 4-mg/cm² ^9Be target. ^{37}Ca was selected by FRS and was incident on a 700-mg/cm² secondary ^9Be target at 195.7 MeV/u. ^{34}Ar was produced via a secondary 2p1n removal reaction. γ rays were detected using the RISING setup consisting of 15 Cluster HPGe detectors, 8 MINIBALL HPGe detectors, and 8 Hector BaF₂ detectors. Fragments were detected using the calorimeter telescope CATE consisting of position sensitive Si ΔE detectors and CsI(Tl) E_{res} detectors. Measured E_γ , I_γ .

 ^{34}Ar Levels

$E(\text{level})$

0
2098 4
3291 5

 $\gamma(^{34}\text{Ar})$

<u>$E_\gamma^\dagger$</u>	<u>$E_i(\text{level})$</u>	<u>E_f</u>	<u>Comments</u>
1193 2	3291	2098	E_γ : 1193 2 from Cluster detectors and 1193 2 from MINIBALL detectors. Possible contributions from a 1184 γ from ^{35}Ar .
2098 4	2098	0	E_γ : weighted average of 2093 4 from Cluster detectors and 2100 3 from MINIBALL detectors.

[†] From 2007DoZV with only statistical uncertainties from a Gaussian fit on linear background.

 $^9\text{Be}(^{37}\text{Ca}, ^{34}\text{Ar}\gamma)$ 2007DoZVLevel Scheme